

Layout Challenges

Narrow layouts

↑ Execution time

Single wide layout

↓ Utilization

Balanced layouts ✓

↓ Execution time

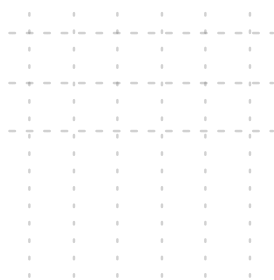
↑ Utilization

Long distance 🚫

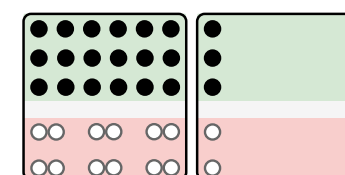
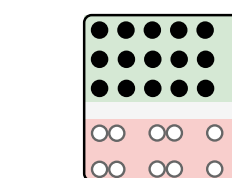
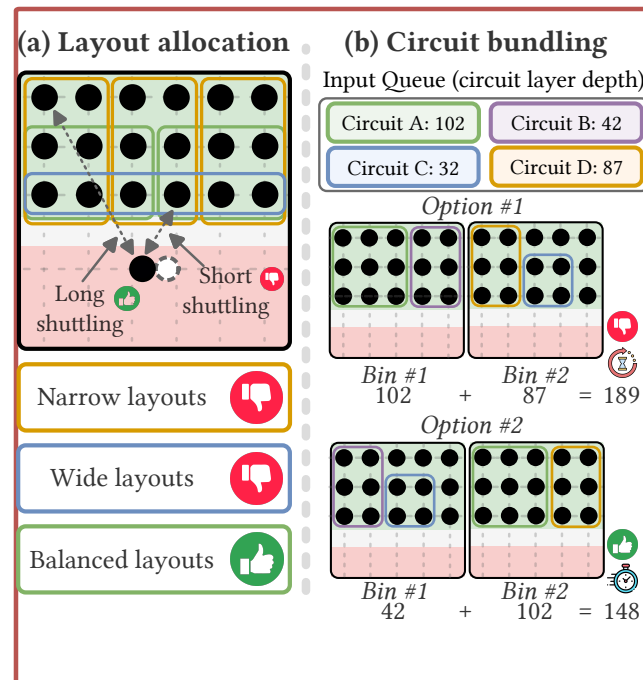
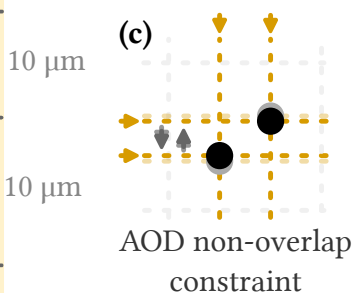
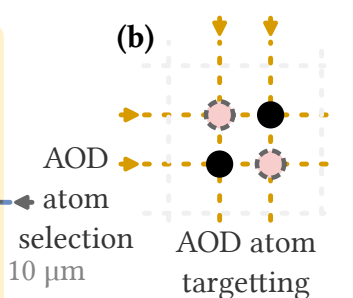
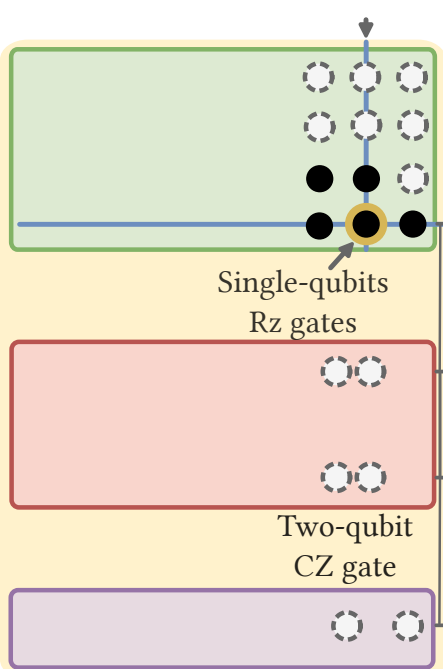
Short distance 👍

- Entanglement Zone

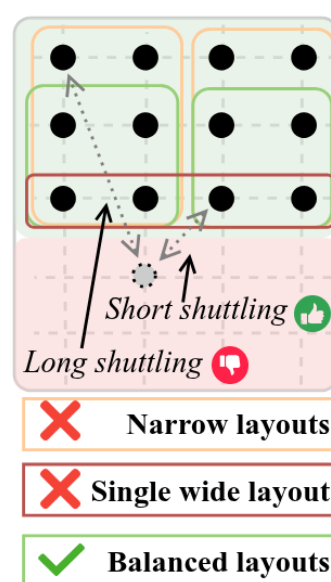
- Storage Zone



Circuit depths (#layers)



(a) Layout allocation



(b) Circuit bundling

